PRODUCT REQUIREMENT DOCUMENT (PRD)

MeshTab: Cross-Currency Group Settlement on Monad

Close the tab. Keep the friends.

October 2026 | Version 1.0

Document Information

Product Name	MeshTab
Team	**Osasumwen Ngozi Scott**: Co-founder and Product Lead (ONSInsights) **Miracle Aigbomian**: Co-founder and Developer (miracles2motion)
Feature Set	MVP: multi-currency expense logging, receipt scan, cross-currency debt netting with visible math, one-tap gasless stablecoin settlement, offline expense drafting, judge self-test path
Target Market	Friend groups that share costs, especially across currencies and borders (trips, remote friends, shared holidays). Secondary: domestic groups such as roommates and event crews.
Event / Track	Monad Metropolis Hackathon (Sept 1 to Oct 13, 2026), Consumer Products & Payments
Network	Monad Testnet (chain ID 10143). Mainnet (chain ID 143) is out of scope for this release.
Live Application	https://mesh-tab.vercel.app
Deployed Contracts (testnet)	MeshTabAgoraLedger: 0x54f02333ea8D80C175A1C8Bdb71ef709cF424C83. Settlement token: the issuer's official testnet AUSD, 0xa9012a055bd4e0eDfF8Ce09f960291C09D5322dC. Funds are claimed from the issuer's official faucet, 0xd236c18D274E54FAccC3dd9DDA4b27965a73ee6C. Addresses to be confirmed on the explorer (V-08, V-16).
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Status	Complete.
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Primary Stakeholders	Metropolis judges, sponsor-bounty reviewers, hackathon team

1. Executive Summary

MeshTab is a group-expense app for friends who share costs, built for groups that spend in more than one currency. It records every expense as a dollar value, nets the whole group's balances into the fewest payments, and lets each debtor pay with one tap and no gas fee. Netting runs off-chain. A small ledger contract on Monad executes the token transfers and emits events. The product is a hybrid of an app server and a blockchain. It is not custodial, not audited, and uses no real money.

Web2 expense trackers record who owes whom, and most net debts only within a single currency. Web2 payment apps move money but do not net a group. Web3 payment apps settle instantly but expect users to manage wallets, seed phrases and gas. On a trip with several currencies, a group ends up with a separate payment list per currency, settled somewhere else. MeshTab closes that gap: one dollar ledger, one netted payment list, one tap to pay.

How to read the numbers in this document. Any figure shown in the demo or in an example is illustrative. Results depend on how many people are in a tab, how many currencies are used, who paid, and how costs were split. The product guarantees properties, not fixed counts:

1. Every tab's balances sum to exactly $0.00.
2. The number of payments never exceeds the number of members with a non-zero balance, minus one.
3. The payments settle each member's net position exactly, to the cent.

Appendix A shows one worked example. Anyone can reproduce these properties with their own numbers in the live app (Section 12).

Evidence status. No primary user research was conducted for this release. Evidence is secondary (Section 2, Appendix B). Premises that are not yet proven are labeled as hypotheses and are tested by the validation plan in Section 12.

Judges can test it themselves. The live app has a sample tab that needs no sign-in, and a testnet mode with a built-in faucet. A guided self-test path (Section 12.1) lets an evaluator reproduce the core claims with their own tab in about five minutes.

MVP Scope Boundary (intentional design choice). This release runs on Monad Testnet, is unaudited, holds no user funds, and does not enforce payment. Settlement uses the issuer's official testnet AUSD, claimed from the issuer's own faucet, so every payment is a real token transfer on testnet with no real value. Fiat on/off-ramps, a virtual card, yield, escrow and mainnet are explicitly deferred (Section 5.2). The Metropolis payments track describes a payments app that never mentions a blockchain to the person using it, so jargon-free UX is treated as a product requirement, not a preference.

Go / No-Go Decision: GO, with a constrained scope and a one-tap-first build mandate. The core loop (log in any currency, net, settle in one tap) ships. Anything that cannot be demonstrated or verified in the live app is labeled as roadmap.

2. Problem Statement

MeshTab addresses six connected problems in shared spending. Evidence is secondary or calculated, as labeled. References (S1 to S8) are in Appendix B.

Pain Point	Segment Affected	Evidence	Evidence Type
Organizers front costs and repayment is slow or incomplete	Organizers, coordinators	A US payments-network consumer survey found 76% of Gen Z who fronted group costs were not fully repaid, and 14% of Gen Z (19% of Millennials) ended a relationship over repayment (S1)	Secondary research (US sample)
Multi-currency tabs leave one payment list per currency	Travelers, nomads, cross-border friends	Web2 expense trackers net debts within each currency separately (S2). On the worked example in Appendix A, this leaves more payments than one cross-currency net	Verified behavior + calculation
Tracking and paying are separate steps	All members	Web2 expense trackers record debts, and where in-app payment exists it is limited to a few countries and bank rails (S3). Web2 payment apps move money but do not net a group	Verified behavior
Cross-border repayment can add fees and FX markups	Cross-border members	Industry observation. Not quantified in this document	Hypothesis
Crypto settlement is hard for non-crypto users (wallets, seed phrases, gas)	Non-crypto members	Design premise drawn from how Web3 payment apps work today. Tested in usability sessions (V-14)	Hypothesis
Unresolved debts cause avoidance and social friction	Debtors, organizers	S1 (relationship impact) plus persona behaviors in Section 4	Secondary research + hypothesis

3. Goals and Success Metrics

Targets are expressed as properties and thresholds that hold for any tab, not as demo-specific counts. Where a target depends on measurement, the measurement method and the evidence required to close it are stated (Section 13).

ID	Type	Metric	Target	Measurement Method
M-01	North Star	Payments per tab vs. a per-currency baseline	Payments never exceed (members with non-zero balance minus 1) in 100% of generated tabs. Median payments saved vs. the per-currency baseline is reported, not pre-asserted	Simulation over randomized tabs (2 to 20 members, 1 to 4 currencies); baseline computed per currency
M-02	Input: Activation	Time to first split (sign in, join or create, log one expense)	Median under 30 seconds	Timed sessions, at least 5 first-time testers
M-03	Input: Completion	Debtors who finish Settle Up once opened	90% or more	Moderated sessions, at least 5 testers (testnet mode)
M-04	Input: Accuracy	Balance invariants hold	100% pass: balances sum to $0.00 and payments never exceed (non-zero members minus 1). Submission run: 1,000 random tabs, passed. To extend to 10,000 tabs before any mainnet use	Property tests (test-netting-properties.mjs): 3 to 20 members, 3 to 17 expenses, uneven splits, five currencies, remainder cents
M-05	Output: Speed	Tap-to-confirmed time for one settlement	Measured: median 1.28 s, fastest 1.19 s, slowest 1.54 s over 10 real testnet settlements on 2 devices (Monad Testnet). Target median 2.0 s or less. Republish if the build changes. Protocol finality reference: 800 ms	Timed runs on 2 devices
M-06	Output: Cost	Gas paid by the user	$0 in 100% of sponsored settlements	Paymaster logs vs. settlement events
M-07	Output: Integrity	Offline operations that end as synced or visibly failed	100% (none silently dropped)	Offline test with at least 50 queued operations
M-08	Output: Comprehension	Testers who correctly explain who pays whom after netting	4 of 5 or better	Moderated sessions
M-09	Output: Self-test	First-time testers who complete the self-test path unaided	4 of 5 or better, each in 5 minutes or less	Observed sessions on the live app
M-10	Output: Trust	Testers who can say what is real and what is not (testnet, test token)	4 of 5 or better	Post-session question

3.1 Analytics Events (no personal data)

Event	Fires when	Properties	Feeds
tab_created	A tab is created	member_count, default_currency, mode	M-02
expense_logged	An expense is saved	currency, split_type, source (manual or scan), offline flag	M-02, M-07
netting_run	Netting is shown	members_nonzero, payments_count, currencies_count	M-01, M-04
settle_opened	Settle Up is opened	payments_owed, authorizations_required	M-03
settle_confirmed	A settlement is confirmed on-chain	tap_to_confirm_ms, sponsored (true/false)	M-03, M-05, M-06
settle_failed	A settlement fails	reason_code	M-03
outbox_flushed /	Offline queue syncs or an operation	ops_synced, ops_failed	M-07

Event	Fires when	Properties	Feeds
outbox_failed	fails		

4. User Personas

MeshTab serves a wider audience than a single niche, so this document uses ten personas, grouped into three tiers by how well the core loop addresses their pain today. Tiers decide where acceptance criteria are written (Section 7) and how claims are made.

4.1 Persona Overview

Persona	Role	Core Need	Key Frustration	Tier
Aisha Bello, 27, Lagos	Senior Product Designer. The Group Fronter	Get reimbursed accurately and promptly without chasing friends	Carries the group's upfront cost, fears seeming petty when she chases, struggles to receive money across borders	Primary
Kemi Adeyemi, 29, London	Strategy Consultant. The Crypto-Curious Expat	Split multi-currency costs with minimal FX loss and settle fast	Groups move gross amounts that netting would cancel. Friends refuse wallets, seed phrases and gas tokens	Primary
Ben Woodward, 26, Toronto	Data Analyst. The Scorekeeping Moderate	Every split verifiably fair to the cent, with the netting math visible	Cannot confirm numbers were not changed. Fees on small payments distort his budget	Primary
Danny Vance, 25, Toronto	Freelance Motion Animator. The Frictionless Debtor	See what he owes and clear it quickly	A tangle of small debts to several people causes avoidance. Cross-border payments take too many steps	Primary
Sarah Jenkins, 31, Lisbon	Remote UX Researcher. The High-Frequency Nomad	Log and settle from her phone in one tap, in one app	Fragmented tools (chat, notes, payments) and slow international clearing	Secondary
Liam Vance, 31, Bali	Remote Full-Stack Engineer. The Digital Nomad	Split lodging, coworking and rentals across countries without FX loss	Separate payment lists per currency. Non-crypto friends reject crypto settlement	Secondary
Tunde Adebayo, 26, Lagos	Digital Marketing Specialist. The Super-Social Coordinator	Instant reimbursement without reminder messages	Fronts the card for friends and reconciles bank alerts against his ledger by hand	Secondary
Chloe Jenkins, 22, New York	Student and Freelance Designer. The Payment Avoider	Track shared apartment costs with roommates without dread	Bill-splitting anxiety. Wallets, seed phrases and gas feel frightening	Adjacent
Alex Chen, 24, Taipei	Web3 Developer. The Onchain Native	Settle real-world debts in stablecoins and split conference costs with his team	Paying gas for a small dinner. Teammates sending tokens in circles	Adjacent
Elena Rostova, 35, Berlin	Property Manager. The Shared-Economy Pragmatist	Shared costs balanced to the cent, with clear boundaries	Circular A to B to C to A payments. Distrusts manual inputs	Adjacent

4.2 Tier Definitions

Tier	Definition	Commitment in this PRD
Primary	The core loop (log in any currency, net, settle in one tap) solves their main pain, and they appear in the demo benchmark	Dedicated acceptance criteria, usability testing, and metrics (Section 7, Section 12)
Secondary	The core loop helps, but the edge is smaller (for example single-currency tabs) or depends on roadmap items such as a fiat off-	Covered by the same stories and criteria. No separate commitments

Tier	Definition	Commitment in this PRD
	ramp	
Adjacent	The need is only partly met or not met today. They are acknowledged so the product does not over-promise	No requirements are promised. Gaps are stated openly in Section 4.3

4.3 What MeshTab Delivers, and What It Does Not

Persona	Delivered in this release	Not delivered, stated openly
Aisha	One-tap payback into her wallet, netted across currencies	An in-app nudge exists, but no push or email reminders and no enforcement. Nothing forces a debtor to pay
Kemi	Cross-currency netting. Friends sign in with a passkey and need no wallet	Three currencies use indicative rates. The token is a testnet token with no value
Ben W.	Per-member net positions and a before/after graph show the math. Each payment is verifiable on the explorer	Expenses are stored off-chain, so retroactive edits are not provably prevented. Edits are logged instead
Danny	Fewer payments and one-tap settlement	Converting the stablecoin to local currency is roadmap
Sarah	Passkey sign-in, real-time logging, receipt scan	Same off-ramp gap
Liam	Multi-currency netting	Same off-ramp gap
Tunde	One-tap payback	On single-currency tabs the payment count matches other tools, so the edge is settlement only
Chloe	No wallet, seed phrase or gas for the user	Domestic roommate use has little count advantage. Only an in-app nudge, no push or email reminders
Alex	Sponsored, batched payment calls	Wants self-custody and mainnet settlement, both out of scope
Elena	Netting removes circular payments	An unalterable ledger is not delivered. The system is a hybrid

5. Scope

5.1 In Scope: Testnet MVP

Feature Area	Details
Passwordless authentication	Passkey (FaceID or TouchID), Google, or email through Privy. A smart wallet is created automatically for every user
Tabs and members	Create a tab, join by link or 8-character code, manage members. A joining member's smart-account address is registered in the roster automatically. Up to 15 members per tab are supported, with a hard ceiling of 20
Expense logging	Manual entry with equal, exact, percentage and shares splits. Integer-cent math with deterministic remainder distribution. Edit and delete with automatic recalculation
Receipt scan	Gemini Flash reads total and merchant into a draft the user must confirm
Multi-currency	EUR, GBP, JPY, CAD, AUD and CHF at live Pyth Hermes rates. SGD, AED and NGN at indicative, clearly labeled rates. USD value fixed at entry
Netting	Greedy net-position matching with at most (non-zero members minus 1) payments, per-member net positions, before/after graph, and a reconciliation indicator
One-tap settlement	The debtor taps Confirm Settlement, then confirms on the device: a biometric check (Face ID, Touch ID or Windows Hello) when a passkey is enrolled, or a slide or typed-amount confirmation when not. One user operation batches the token approval and every payment the debtor owes. Gas is sponsored (Pimlico paymaster) for whitelisted contracts. An explorer link is shown after
Test funds	Add Funds claims testnet AUSD from the issuer's official faucet: 10,000 per claim, a 100,000 wallet

Feature Area	Details
	ceiling, and one claim per 60 seconds
Offline drafting	Log expenses and recompute netting offline. Sync when reconnected
Mode separation	Sample mode (local simulation, no chain writes) and Testnet mode (real transactions), with a persistent banner
Judge self-test path	Sample tab reachable without sign-in. A guided path from landing to a verified settlement

5.2 Out of Scope

Out of Scope	Rationale
Mainnet and the production stablecoin	The token address is fixed in the ledger's constructor, so mainnet means a new deployment, plus a funded paymaster, a security review and a real-money compliance review. It is not a configuration change
Fiat on-ramp, off-ramp, and debit-card funding	Testnet cannot support provider sandboxes that run on mainnet assets. Planned through Privy's funding features (Stripe on-ramp), subject to provider coverage of networks, assets and currencies
Virtual card and yield vault	Prototype UI only. A virtual card is planned through Privy's card program, subject to program access and supported assets. The contract holds no tab funds, so a vault needs a custody design first
Group-atomic settlement in one transaction	The contract moves one payer-to-recipient transfer per call. No member can move another member's funds
Payment enforcement and escrow	Nothing forces a debtor to pay. Escrow is roadmap
Offline settlement	A chain transaction cannot be broadcast without a connection
Push, email and SMS reminders	Not built. An in-app Ping All nudge exists. The active tab is polled every 5 seconds and balances update on refresh
Security review and audit	The prototype is unaudited. A pre-mainnet review (an independent audit, plus a security scan if selected for the ack3 sponsor prize) precedes any mainnet launch

6. Functional Requirements

Requirements describe behavior that holds for any tab. Where a requirement depends on a build check, it is listed in the Verification Register (Section 13). Personas are listed by first name.

ID	Feature	Requirement Description	Priority	Personas
FR-01	Passwordless sign-in	Users sign in with a passkey, Google, or email. A first-time user reaches the home screen without entering a password, installing an extension, or seeing a seed phrase.	Must	All
FR-02	Smart wallet	An ERC-4337 smart account is created automatically for each authenticated user. No manual wallet setup is shown.	Must	All
FR-03	Profile onboarding	First-time users set a display name and avatar in under 30 seconds.	Should	Chloe, Danny
FR-04	Create tab	Users create a tab with a name, a default currency from the supported list, an optional category, and initial members.	Must	Aisha, Tunde
FR-05	Invite and join	Each tab has an 8-character join code and a deep link with a 30-day rolling expiry. Opening the link signed-out leads through sign-in into the tab.	Must	Danny, Chloe
FR-06	Members and wallet linking	A member's smart-account address is registered in the roster automatically when they join by link or code. A member added by name only has no address until they join, and the app blocks settlement to them (Account not linked).	Must	Danny, Aisha
FR-07	Manual expense and splits	Users enter amount, description, payer, participants, category and date. Split types: equal, exact, percentage, shares. Splits that do not total the	Must	Ben, Elena

ID	Feature	Requirement Description	Priority	Personas
		amount or 100% are blocked with the difference shown.		
FR-08	Integer-cent math	All money is stored as integer cents. Equal splits that do not divide evenly distribute remainder cents deterministically: the same inputs always give the same result. Balances always sum to $0.00.	Must	Ben, Elena, Kemi
FR-09	Edit and delete expenses	Editing or deleting an expense recalculates all balances and netting. A settlement is stored as a separate payment record and is never silently rewritten.	Must	Ben, Elena
FR-10	Change history	Each expense shows who created or last changed it and when.	Should	Ben, Elena
FR-11	Multi-currency capture	Expenses can be entered in EUR, GBP, JPY, CAD, AUD, CHF, SGD, AED or NGN. The USD value is calculated at entry and stored in cents. Later rate changes never alter an existing expense. The original currency and amount remain visible.	Must	Kemi, Liam
FR-12	Rate labeling	EUR, GBP, JPY, CAD, AUD and CHF use live Pyth Hermes rates. SGD, AED and NGN use indicative hardcoded rates and show an 'indicative' badge. If a live feed is unavailable, the indicative rate is used and flagged.	Must	Kemi, Liam
FR-13	Receipt scan	A camera or upload sends the image to the scan endpoint, which returns a draft total and merchant. The draft is never saved without user confirmation and is always editable.	Should	Sarah, Tunde
FR-14	Netting	The system recomputes every net position after any change and produces a payment list using greedy net-position matching. The list contains at most (members with non-zero balance minus 1) payments and settles each net position exactly. The result is not guaranteed to be the theoretical minimum.	Must	Kemi, Ben, Elena
FR-15	Show the math	The tab shows each member's net position, the payment list, a before/after graph, and a reconciliation indicator that displays the total of all balances as $0.00.	Must	Ben, Kemi, Elena
FR-16	Settle Up	A debtor sees every payment they owe with amount and recipient name, and the total before starting. One tap on Confirm Settlement starts a single operation that batches the token approval and every payment the debtor owes. The device then asks for confirmation: a biometric check (Face ID, Touch ID or Windows Hello) when a passkey is enrolled, or a slide or typed-amount confirmation when not.	Must	Danny, Aisha
FR-17	Sponsored gas	The user never needs a gas token. A paymaster sponsors testnet user operations. Sponsorship covers whitelisted contracts only and is bounded by a monthly policy spend ceiling with an alert. There is no per-user cap on testnet.	Must	Danny, Alex
FR-18	Settlement history	Every settlement shows payer, recipient, amount, date and status, with an explorer link behind 'View receipt'. Members cannot edit settlement records.	Should	Ben, Elena
FR-19	Creditor view	When a payment is confirmed on-chain, the creditor's tab balance and history update on the next load or refresh.	Should	Aisha, Tunde
FR-20	Mode separation	Sample mode never writes on-chain and uses its own storage namespace. Testnet mode writes on-chain. A persistent banner shows the active mode. Clearing sample data never touches testnet data. Three workspace modes exist (sample, blank, testnet), set in Settings. A separate Developer and Web3 toggle (Settings, Security) exposes raw wallet modals and a private-key export. Judges land in sample mode with the Lisbon trip pre-filled, in the consumer view.	Must	Evaluators
FR-21	Faucet	In testnet mode, Add Funds claims testnet AUSD from the issuer's official faucet (10,000 per claim, 100,000 wallet ceiling, one claim per 60 seconds). The claim's gas is sponsored. Insufficient funds at Settle Up leads to the claim flow. A claim history lists earlier claims with explorer links.	Must	Evaluators, Alex
FR-22	Offline drafting	Expense changes queue in a local outbox with optimistic UI and sync in order on reconnect. Conflicts resolve last-write-wins on server arrival, with	Should	Sarah, Liam

ID	Feature	Requirement Description	Priority	Personas
		the change attributed. Settle Up is disabled offline.		
FR-23	Active-balance guard	A member with a non-zero balance cannot be removed. The message states the balance.	Should	Aisha
FR-24	Disclosure	The app states 'Unaudited testnet prototype. Not real money.' and labels the token as testnet AUSD with no value. Roadmap features are labeled as not live. The Cash Out and Virtual Card screens are simulations and say so (Cash Out shows Testnet Payout Simulated). No UI text claims an audit, custody, or a trustless ledger.	Must	Evaluators, Chloe
FR-25	Judge self-test path	The landing page links to the sample tab in one tap without sign-in. A guided path (Section 12.1) leads from the sample to a verified testnet settlement.	Should	Evaluators
FR-26	Auto-lock and cues	A soft lock after 5 minutes of inactivity keeps the session and offline queue. Sound and haptic cues confirm success.	Could	Sarah
FR-27	Tab capacity	Up to 15 members per tab are supported and tested. The hard ceiling is 20. Adding a 21st member is blocked with a clear message.	Should	All
FR-28	In-app nudge	A creditor can send an in-app reminder to one debtor or to all debtors (Ping All). It appears as a notification and as an item in the tab's activity feed, with a settlement link. Creditors never nudge themselves. There is no email, SMS or push.	Could	Aisha, Tunde

7. User Stories with Acceptance Criteria

All stories follow the INVEST principle. Criteria are numbered, measurable, and independent of any particular demo figures. Example amounts, where they appear, are illustrative and labeled 'e.g.' A story is done only when every criterion passes by the stated test method and the Definition of Done below is met.

ID	Epic and Persona	User Story	Acceptance Criteria
US-01	Authentication Aisha, Chloe, Sarah	As a new member, I want to sign in with my face or fingerprint so that I never deal with a wallet.	1. Sign-in offers passkey, Google and email. 2. First sign-in never asks for a password, extension, seed phrase or gas token. 3. A smart account is created in the background and the home screen appears within 10 seconds on a 4G connection. 4. A failed sign-in shows a plain-language message and a retry. No error codes are shown. 5. A returning user is restored without re-entering details. **Test:** Manual run on 3 devices (iOS, Android, desktop) plus timing.
US-02	Create and invite Aisha, Tunde	As an organizer, I want to create a tab and share a link so that friends join in seconds.	1. Name and default currency are required. Category is optional. 2. A share link and a 6-character code are generated at creation. 3. The creator appears as the first member. 4. The link stays valid for 30 days from last use. 5. The creator can add members by name only. **Test:** Functional test plus link-expiry test.
US-03	Join Danny, Chloe	As an invited friend, I want to open a link and land in the tab so that I can start without setup.	1. Opening the link signed-out leads through sign-in and then into the tab. 2. The joiner appears once in the roster, even if the link is opened twice. 3. An invalid or expired link shows a message explaining how to get a new one. 4. The joiner sees existing expenses and balances immediately. **Test:** Functional test with two accounts.
US-04	Expenses and splits Ben, Elena	As a member, I want to log who paid and how it is split so that every balance is fair to the cent.	1. Amount, description, payer, participants, category and date are accepted. 2. Equal, exact, percentage and shares splits are supported.

ID	Epic and Persona	User Story	Acceptance Criteria
			3. Exact or percentage splits that do not total the amount (or 100%) are blocked and the difference is shown. 4. Equal splits that do not divide evenly distribute remainder cents deterministically. 5. After saving, balances across the tab sum to exactly $0.00, for any number of participants (M-04). **Test:** Unit and property tests over at least 10,000 random tabs.
US-05	Foreign currency Kemi, Liam	As a traveler, I want to log an expense in euros or pounds so that the group sees one dollar value.	1. The currency picker offers the 9 supported currencies. 2. Live rates apply to 6 currencies. An 'indicative' badge shows on the other 3. 3. The USD value is fixed at entry and stored in integer cents. Later rate changes never alter it. 4. The original currency and amount remain visible beside the USD value. 5. If a live feed is unavailable, the indicative rate is used and the expense is flagged. **Test:** Functional test with a mocked rate feed, plus a stored-value check after a rate change.
US-06	Receipt scan Sarah, Tunde	As a member, I want to photograph a receipt so that I do not type totals.	1. Camera or upload sends the image for extraction. 2. A draft with total and merchant is shown and is never saved automatically. 3. The draft is editable before saving. 4. On failure or low confidence, manual entry opens with the amount blank. 5. The scan endpoint is rate-limited, and the image is not retained beyond the request. **Test:** Timed run over 10 receipts, rate-limit load test, and code review.
US-07	Netting and the math Ben, Kemi, Elena	As a member, I want one clear list of who pays whom, and to see how it was calculated, so that I can trust it.	1. The tab shows each member's net position (positive or negative) and the payment list. 2. The number of payments never exceeds (members with non-zero balance minus 1). 3. The payments settle every net position exactly, to the cent. 4. A before/after graph is available. 5. The result recalculates after every add, edit or delete, and the same inputs always give the same list. 6. A reconciliation indicator shows the total of all balances as $0.00. 7. Netting works offline from local data. **Test:** Property tests (M-01, M-04) plus a manual run with user-chosen numbers (Section 12.1).
US-08	One-tap settlement Danny, Aisha	As a debtor, I want to pay in one tap so that I am done in seconds.	1. Settle Up lists every payment owed with amount and recipient name. 2. One tap on Confirm Settlement starts one operation that batches the approval and every payment owed. The total is shown before starting. 3. No gas fee or token is shown or required. 4. On success the debtor's balance falls by exactly the amount paid, and an explorer link is available behind 'View receipt'. 5. On failure the balance is unchanged and no partial state remains. 6. Tap-to-confirmed time is measured and published (M-05). 7. The device then asks for confirmation: a biometric check when a passkey is enrolled, or a slide or typed-amount confirmation when not. Cancelling sends nothing. **Test:** 10 timed testnet settlements on 2 devices, plus a forced-failure test.
US-09	Receiving payment Aisha, Tunde	As a creditor, I want to see when I have been paid so that I stop chasing.	1. After a confirmed payment, my balance updates on the next load or refresh. 2. The payment appears in history with payer, amount and time. 3. My test-token balance reflects the payment. **Test:** Two-account functional test.

ID	Epic and Persona	User Story	Acceptance Criteria
US-10	Missing wallet Danny	As a debtor, I want a clear message when a friend cannot be paid yet so that I know what to do.	1. Settle Up to a member without a linked wallet is blocked. 2. The message names the person and the next step. 3. No transaction is sent. 4. Other payments in the list are unaffected. **Test:** Functional test with a name-only member.
US-11	Test funds Evaluators, Alex	As a tester, I want free test tokens so that I can complete a settlement.	1. One tap claims testnet AUSD from the issuer's official faucet (10,000 per claim) in testnet mode only. 2. The balance updates visibly within the confirmation time. 3. A claim within 60 seconds of the last, or one that would exceed the wallet ceiling, shows a clear reason and when to retry (EC-19). 4. Insufficient funds at Settle Up leads to the claim flow. 5. The token is labeled testnet AUSD with no value. **Test:** Functional test, new wallet and repeat wallet.
US-12	Offline drafting Sarah, Liam	As a traveler with poor signal, I want to log expenses offline so that I never lose them.	1. An offline banner appears within 5 seconds of losing connection. 2. An expense saves locally and shows 'waiting to sync'. 3. On reconnect, queued changes sync automatically and in order. 4. A change that cannot sync is shown as failed with a retry, and is never dropped (M-07). 5. Settle Up is disabled offline, with an explanation. 6. The queue survives an app restart. 7. Concurrent edits resolve last-write-wins and show who changed the expense. **Test:** Offline test with at least 50 queued operations, kill-and-relaunch test.
US-13	Edit and delete Ben, Elena	As a member, I want to correct a mistake so that balances stay right.	1. Editing amount, payer or split recalculates balances and netting. 2. Deleting an expense removes it and recalculates. 3. Each expense shows who changed it last and when. 4. A settlement made before an edit stays as a separate payment line and is not rewritten. **Test:** Functional test plus property test after random edits.
US-14	Settlement history Elena, Ben	As a member, I want a record of every payment so that I can audit the tab.	1. Each settlement shows payer, recipient, amount, date and status. 2. An explorer link is available for each confirmed settlement. 3. Members cannot edit or delete settlement records. **Test:** Functional test.
US-15	Member removal Aisha	As an organizer, I want to remove a member without losing money owed.	1. A member with a non-zero balance cannot be removed, and the message states the balance. 2. A zero-balance member can be removed and history is preserved. **Test:** Functional test.
US-16	Mode clarity Evaluators	As an evaluator, I want to know whether I am in a sample or a real testnet tab.	1. A persistent banner shows Sample or Testnet on every screen. 2. Sample mode never produces a transaction. 3. Switching modes never mixes data, and clearing sample data leaves testnet data untouched. **Test:** Functional test plus storage-key inspection.
US-17	Disclosure Evaluators, Chloe	As an evaluator, I want the app to say plainly what is and is not real.	1. The footer or About screen states 'Unaudited testnet prototype. Not real money.' 2. The token is labeled testnet AUSD with no value. 3. Roadmap features are labeled as not live. 4. No UI text claims an audit, custody, or a trustless ledger (M-10). **Test:** Copy audit of all screens (V-13).
US-18	Judge self-test Evaluators	As a judge, I want to test the app with my own numbers so that I can verify the claims myself.	1. The landing page links to the sample tab in one tap, with no sign-in. 2. A first-time user completes the self-test path in 5 minutes or less (M-09). 3. A tab with at least 3 members and 2 currencies produces a netting

ID	Epic and Persona	User Story	Acceptance Criteria
			result that obeys the properties in Section 1. 4. Every step in the path shows a visible success state. 5. Judges can enter any figures. Results are never hard-coded to the demo. **Test:** Observed sessions with 3 first-time users on 2 devices (V-14).

7.1 Definition of Done

- Every acceptance criterion passes by its stated test method, with evidence stored in the evidence pack.
- Property tests (M-01, M-04) pass on the final build.
- No retired phrase or unverified figure appears in the UI, README, demo script or this document (V-13).
- The feature works on a 360 px mobile viewport and in both light and dark modes.
- Failure paths in Section 11 are exercised, not only the happy path.

8. Prioritization (MoSCoW)

MoSCoW is used because it forces an explicit statement of what will not be built, which matters here: this is a time-boxed hackathon release, and over-promising is a larger risk than under-delivering.

Priority	Feature Area	Rationale	Stories
Must	Passkey sign-in and smart wallet	Without frictionless entry the non-crypto premise fails.	US-01
Must	Tabs, invite, join, wallet linking	Defines the group every other feature depends on.	US-02, 03, 10
Must	Expense logging with integer-cent math	Trust in the numbers is the foundation of the product.	US-04, 13
Must	Multi-currency capture and labeling	This is the differentiator against per-currency netting.	US-05
Must	Netting with visible math	The core value proposition and the thing judges will probe.	US-07
Must	One-tap sponsored settlement	The only part that moves value and the demo's hero moment.	US-08
Must	Mode separation, faucet, disclosure	Lets evaluators test safely and keeps every claim honest.	US-11, 16, 17
Should	Receipt scan	Speeds logging but is not required for the core loop.	US-06
Should	Offline drafting	Useful, but offline entry is common in this category, so it is not a differentiator.	US-12
Should	Settlement history, creditor view, change history	Builds trust and supports audit-minded users.	US-09, 14
Should	Judge self-test path	Turns claims into something verifiable in minutes.	US-18
Could	Auto-lock, sound and haptic cues, member-removal guard polish	Quality touches that do not change the core loop.	US-15
Won't (MVP)	Mainnet, fiat ramps, card, yield, escrow, push, email and SMS reminders, offline settlement, group-atomic settlement	See Section 5.2. Each needs a dependency, a custody decision or a compliance review that this release does not include.	None

9. Traceability Matrix

Every persona connects to at least one story, requirement and metric. Adjacent personas are traced to the criteria they rely on, but no additional commitments are made for them.

Persona (Tier)	Stories	Requirements	NFRs	Metrics
Aisha (Primary)	US-01, 02, 08, 09, 15	FR-01, 04, 05, 06, 16, 19, 23	P02, S02	M-02, M-03
Kemi (Primary)	US-05, 07, 08	FR-11, 12, 14, 15	P04, R02	M-01, M-04
Ben (Primary)	US-04, 07, 13, 14	FR-07, 08, 09, 10, 15, 18	R02, S06	M-04, M-08
Danny (Primary)	US-03, 08, 10	FR-05, 06, 16, 17	P02, A04	M-03, M-05
Sarah (Secondary)	US-01, 06, 12	FR-01, 13, 22, 26	P03, R01	M-07
Liam (Secondary)	US-05, 12	FR-11, 12, 22	R01	M-01, M-07
Tunde (Secondary)	US-02, 06, 08, 09	FR-04, 13, 16, 19	P02	M-03, M-05
Chloe (Adjacent)	US-01, 03, 17	FR-01, 02, 03, 24	A04	M-10
Alex (Adjacent)	US-08, 11	FR-17, 21	S08	M-06
Elena (Adjacent)	US-04, 07, 13, 14	FR-08, 09, 15, 18	R02	M-04

10. Non-Functional Requirements

Category	ID	Requirement	Acceptance Criteria / Metric
Performance	NFR-P01	Home screen first contentful paint is 1.2 seconds or less on mobile 4G.	Lighthouse and a real-device run
Performance	NFR-P02	Settlement: protocol finality is 800 ms (published by Monad). Tap-to-confirmed time is measured and published. Measured on testnet: median 1.28 s, slowest 1.54 s (10 runs, 2 devices).	10 timed testnet settlements on 2 devices. Median and slowest reported (M-05)
Performance	NFR-P03	Receipt scan returns a draft in 2.0 seconds or less (median).	Timed over 10 receipts
Performance	NFR-P04	Netting recompute takes under 100 ms for 20 members and 200 expenses.	Unit benchmark
Performance	NFR-P05	An offline queue of up to 50 operations syncs within 10 seconds of reconnect.	Offline test
Security	NFR-S01	All traffic uses TLS. There is no HTTP fallback.	Host check
Security	NFR-S02	All endpoints that change data require a valid Privy token.	Calls without a token are rejected
Security	NFR-S03	The receipt-scan endpoint is rate-limited to 10 requests a minute per IP (in memory, best effort on serverless hosting). The FX endpoint is public and unauthenticated. Requests beyond the limit receive 429.	Load test (V-09)
Security	NFR-S04	Tab membership is verified before any tab data is returned. A non-member cannot read another tab.	Cross-tab access test
Security	NFR-S05	No private keys or secrets appear in client bundles or the public repository.	Repo and bundle scan
Security	NFR-S06	The ledger contract moves only the caller's own tokens to the named recipient and cannot move another member's funds (permissionless: the caller pays from their own approved balance).	Contract review and test (V-08)
Security	NFR-S07	Money is stored as integer cents. The Lua scripts apply integer -cent values with HINCRBYFLOAT. Moving to HINCRBY is a mainnet-readiness item.	Code review
Security	NFR-S08	Gas sponsorship is restricted to whitelisted contracts and bounded by a monthly policy spend ceiling with an alert at 80%. There is no per-user cap on testnet.	Paymaster policy check (V-09)
Privacy	NFR-PR01	On-chain data is limited to wallet addresses, amounts and a plaintext tab identifier (not hashed). Tab names, descriptions and receipts are never on-chain.	Event inspection on the explorer (V-08)

Category	ID	Requirement	Acceptance Criteria / Metric
Privacy	NFR-PR02	Receipt images are used only for extraction and are not retained beyond the request.	Code review (V-15)
Privacy	NFR-PR03	Users can request deletion of off-chain data. On-chain records are immutable, and this is disclosed.	Documented process
Accessibility	NFR-A01	Tap targets are at least 44 by 44 px. The layout works from 360 px width.	Automated scan and manual check
Accessibility	NFR-A02	Text contrast meets WCAG 2.1 AA. Light and dark modes are supported.	Contrast checker on all screens
Accessibility	NFR-A03	Icon-only buttons have labels. Core flows work at 200% system font size.	Screen reader and font-size test
Accessibility	NFR-A04	Primary flows use plain language with no blockchain terms (wallet, gas, hash). Technical details sit behind 'View receipt'.	Copy audit (V-13)
Reliability	NFR-R01	The offline queue survives an app crash or restart and syncs on next launch.	Kill-and-relaunch test
Reliability	NFR-R02	Balances sum to $0.00 for all inputs, including remainder-cent splits and many members.	Property tests (M-04)
Reliability	NFR-R03	The app stays available through the judging period.	External uptime probe
Observability	NFR-O01	Analytics events (Section 3.1) contain no personal data. Error logs never contain secrets.	Log review
Compatibility	NFR-C01	Works on the latest two versions of Chrome, Safari, Edge and Firefox, on mobile and desktop. Passkey sign-in requires a passkey-capable device, and Google and email are the fallbacks.	Cross-browser run

11. Edge Cases and Error States

ID	Scenario	Trigger	Expected Behaviour	Message
EC-01	Recipient has no wallet	Debtor taps Settle Up	Block. Send nothing. Other payments are unaffected	'[Name] has not set up payments yet. Ask them to open MeshTab.'
EC-02	Not enough funds	Balance is below the amount	Stop before signing. In testnet mode, offer the faucet	'You need $X more to pay this.'
EC-03	Sponsorship rejected or rate-limited	Paymaster refuses	Offer retry. The user is never charged or asked for gas	'Could not process right now. Try again in a moment.'
EC-04	Confirmation is slow	No receipt before the timeout	Show pending, block a second submit, check status by transaction hash	'Still confirming. Please do not tap again.'
EC-05	Tab changes mid-settlement	Another member logs an expense while Settle Up is open	Recompute and ask the debtor to re-confirm the new amount	'This tab just changed. Review the new amount.'
EC-06	Partial transaction	Approval succeeded but payment failed	Not possible inside one user operation. If ever sent separately, a retry reuses the allowance and no balance changes	'Payment did not go through. Try again.'
EC-07	Offline at Settle Up	No connection	Disable settlement, keep logging	'Connect to the internet to pay.'
EC-08	Rate feed unavailable	Live feed down, or a currency without a live feed	Use the indicative rate and flag the expense	'Estimated rate. Not live.'
EC-09	Receipt scan fails or is wrong	Low confidence or quota reached	Open manual entry. Never auto-save	'Could not read this receipt. Enter it manually.'

ID	Scenario	Trigger	Expected Behaviour	Message
EC-10	Uneven cents	Equal split of an indivisible amount	Distribute remainder cents deterministically. The sum stays exact	None
EC-11	Concurrent offline edits	Two members edit the same expense offline	Last write wins on server arrival. Show who changed it	'Updated by [name].'
EC-12	Sample mistaken for real	User expects a chain result in sample mode	Persistent banner. No transaction in sample mode	'Sample tab. Nothing is sent.'
EC-13	Zero or single-member tab	Netting on a tab with no debts, or one member	Show 'All settled' or an invite prompt. No payments listed	'Everyone is settled up.'
EC-14	Single-currency tab	All expenses in one currency	Netting works normally. No count advantage is claimed over other tools	None
EC-15	Payer is not a participant	Expense paid by someone not in the split	Allowed. Payer is credited the full amount and participants are charged their shares	None
EC-16	Edit after settlement	An expense that predates a settlement is edited	Balances recalculate. The settlement remains a separate payment line	'Balance updated. Your earlier payment is unchanged.'
EC-17	Double tap	User taps Settle Up twice	Second tap is ignored while one is in progress	None
EC-18	Tab at capacity	A 21st member is added or joins by link	Block the join and keep existing members unaffected. Tabs of up to 15 are supported and tested, and 20 is the hard ceiling	'This tab is full (20 members).'
EC-19	Faucet limit or outage	Claim within 60 seconds of the last, a wallet near the ceiling, or the faucet is unavailable	Show the reason and when to retry. Settle Up is unaffected	'You can claim again in a minute.' or 'You already have plenty of test funds.'
EC-20	Biometric prompt cancelled	User cancels the Face ID, Touch ID or Windows Hello prompt	Nothing is sent and the balance is unchanged	'Payment cancelled.'

12. Submission, Judge Self-Test and Validation Plan

12.1 Judge Self-Test Path

Evaluators can reproduce the core claims with their own numbers. The path uses the live app and takes about five minutes.

Step	Action	Expected Result
1	Open the live link on a phone or desktop	Landing page loads with a one-tap link to the sample tab
2	Open the sample tab and run netting	Net positions, payment list and before/after graph appear. Reconciliation shows $0.00. Banner says Sample
3	Create your own tab. Add at least 3 members. Log at least 3 expenses in at least 2 currencies, paid by different people	Balances update after each expense. Indicative-rate badges appear where applicable. Reconciliation stays $0.00
4	Run netting on your tab	Payments are at most (members with a non-zero balance minus 1). Payments settle every net position exactly. Results are your own numbers, not demo figures
5	In Settings (Workspace Sandbox Mode) or from the landing page, switch to testnet mode. Sign in with a passkey and tap Add Funds	Banner shows Testnet. Balance shows 10,000 testnet AUSD
6	Settle one payment. The recipient needs a linked wallet, so	One tap on Confirm Settlement, then the device biometric check

Step	Action	Expected Result
	use a second browser profile or device as another member	or a slide or typed-amount confirmation. Success state. Explorer link opens the transaction
7	Open settlement history	The settlement appears. The debtor's balance fell by exactly the amount paid

12.2 Validation Plan

Element	Details
Moderated sessions	At least 5 first-time testers on the live app. Measure time to first split (M-02), Settle Up completion (M-03), comprehension (M-08), self-test completion (M-09) and trust (M-10)
Property testing	At least 10,000 random tabs (2 to 20 members, 1 to 4 currencies, uneven splits). Verify M-01 and M-04
Timing	10 real testnet settlements on 2 devices, and 10 receipt scans. Publish median and p90
Evidence pack	Property-test log, timing sheet, explorer links, the written benchmark output, and the baseline comparison (V-02)
Judging lens (team reading, confirm on the event page)	Does it work end-to-end, UX quality, and Monad integration (V-07)

12.3 Submission Gate

- All Verification Register items marked Must are closed (Section 13).
- Seed benchmark reproduced in the app, and the baseline comparison captured with evidence.
- Rate limit and sponsorship cap live. Sample-mode storage prefixed. Fictional model names removed.
- Copy audit complete: no retired phrases and no unverified claims in the app, README, demo script or this PRD.
- At least one real settlement confirmed on the testnet explorer, with timing recorded.
- Self-test path run by 3 first-time users on 2 devices.

12.4 Post-Hackathon Direction

Mainnet with the production stablecoin and a pre-mainnet security review (an independent audit, plus a security scan if we are selected for the ack3 sponsor prize). Fiat funding and cash-out through Privy (Stripe on-ramp), a virtual debit card through Privy's card program, and local-currency cash-out starting with Nigeria, each subject to provider coverage and program access. Escrow, opt-in session keys, and a group vault with a defined custody model. Monetization is not defined in this release.

13. Verification Register and Open Questions

Each item names the evidence that closes it. Items marked Must block submission. The deadline is October 13, 2026.

ID	Item	Evidence to Close	Owner	Due	Status	Gate
V-01	Benchmark seed in the app reproduces the expected netting output	Netting screen shows $391.54 moving and three payments (seen by the product lead)	Eng	Oct 9	Done	Must
V-02	Baseline comparison for the benchmark in a Web2 expense tracker with debt simplification on	Written per-currency payment list (internal evidence)	PM	Oct 9	Open	Must

ID	Item	Evidence to Close	Owner	Due	Status	Gate
V-03	Property tests for M-01 and M-04	Script test-netting-properties.mjs: 1,000 random tabs, balances sum to $0.00 and payments within the bound, 1,000 of 1,000 passed (dev-run). Extension to 10,000 tabs before mainnet	Eng	Oct 12	Done for submission	Must
V-04	Tap-to-confirmed timing	10-run table: median 1.28 s, fastest 1.19 s, slowest 1.54 s (dev-measured by stopwatch, 2 devices, single-payment batched flow). Live-app hashes: faucet claim 0x7a84c8a2…153f, batched settlement 0xb3e1987d…58c1	Eng	Oct 10	Closed (dev-measured)	Must
V-05	Receipt-scan timing	10-receipt table	Eng	Oct 10	Open	Should
V-06	Does a debtor with several creditors settle in one operation?	Code reading (settlement-drawer.tsx, lines 261 to 290): one operation batches the approval and every creditor payment. Not separately timed	Eng	Oct 12	Closed (dev code reading)	Should
V-07	Metropolis rules and sponsor bounty text: network and token requirements	Quote from the event page	PM	Oct 8	Open	Must
V-08	Contract review of the ledger: caller-only token movement, opaque tab identifier, address on explorer	Code review and explorer link	Eng	Oct 9	Open	Must
V-09	Rate limit on public endpoints and sponsorship guardrails	OCR limit of 10 scans a minute per IP, contract whitelist, and a monthly spend ceiling with an alert (dev-reported)	Eng	Oct 10	Done (dev-reported)	Must
V-10	Build status of FR-10 (change history), FR-15 (reconciliation indicator), FR-25 (self-test path)	Screens or a note moving the item to roadmap	Eng	Oct 9	Open	Must
V-11	Sample-mode storage keys prefixed	Key inspection (dev-reported)	Eng	Oct 9	Done (dev-reported)	Should
V-12	Persona ages and roles approved	Sign-off from the team	PM	Oct 9	Open	Should
V-13	Copy audit: retired phrases removed, chain jargon absent from primary flows	Checklist across the app, README and demo script. Landing items confirmed by the dev	PM	Oct 11	Landing done. One in-app string still says instant (code frozen)	Must
V-14	Self-test path run by first-time users	Session notes for 3 users on 2 devices	PM	Oct 11	Open	Must
V-15	Receipt image retention check	Code review	Eng	Oct 10	Open	Should
V-16	Live app runs end to end on the issuer's official testnet AUSD and MeshTabAgoraLedger: fresh-wallet claim, settlement, history	Faucet claim 0x22dc9d3e…2e04, script-executed proof settlement 0xafd413d0…29f0, live-app batched settlement 0xb3e1987d…58c1, and a written read-out of the faucet state (10,000 drip, 100,000 ceiling, 60 s)	Eng	Oct 9	Dev-verified. Links in the evidence pack	Must
OQ-01	Simplest two-account path for judges to test a real settlement	Documented steps in the README and Section 12.1	PM + Eng	Oct 9	Open	Must

14. Dependencies and Risk Register

ID	Dependency or Risk	Likelihood	Impact	Mitigation	Owner
R-01	Monad testnet RPC is slow or down during judging	Medium	High	Pre-recorded demo video as a fallback. Show the status banner. Retry logic with clear messages	Eng

ID	Dependency or Risk	Likelihood	Impact	Mitigation	Owner
R-02	Paymaster funds run out or rate limits trigger	Medium	High	Pre-fund before judging. Whitelisted contracts only. Monthly spend ceiling with an alert at 80%. Monitor balance. Clear failure message (EC-03)	Eng
R-03	Privy outage blocks sign-in	Low	High	Google and email fallbacks. Status check before the demo	Eng
R-04	Gemini quota exhausted or abused through the public endpoint	Medium	Medium	Rate limit (NFR-S03). Manual entry fallback (EC-09)	Eng
R-05	Indicative rates are mistaken for live rates	Medium	Medium	Visible 'indicative' badge (FR-12). Copy audit	PM
R-06	Testnet AUSD is mistaken for real money	Low	Medium	Label as testnet with no value everywhere (FR-24). Disclosure banner	PM
R-13	Devices without an enrolled passkey, and Developer and Sample modes, sign without a biometric prompt (slide or typed-amount confirmation in the consumer view)	Medium	Medium	Acceptable on testnet with no real value. Disclose it. Before any real-money use, require re-authentication or spend limits	Eng
R-12	Issuer faucet unavailable, drained or rate -limited during judging	Medium	High	Pre-claim a reserve of testnet AUSD in a funding account that can send to testers. Clear failure message (EC-19). Test the faucet daily before judging	Eng
R-07	Unaudited contract	Low	Medium	Disclose on-screen. Caller-only transfers (NFR-S06). No custody of funds	Eng
R-08	Claims outrun evidence (benchmark, speed, comparisons)	Medium	High	Every claim traced to the Verification Register. Figures labeled illustrative	PM
R-09	Judges expect real money or mainnet	Medium	Medium	State the testnet scope up front. Roadmap in Section 12.4	PM
R-10	Hybrid architecture questioned (server state plus chain)	High	Medium	Say it plainly. Explain what is on-chain (payments) and off-chain (netting, history) and why	PM
R-11	Real-money launch needs legal and compliance review	High (post-event)	High	Out of scope now. Required before any mainnet release	Legal

15. Roles and Permissions

Role	Can	Cannot
Visitor (signed out)	View the landing page. Open and use the sample tab	Create a tab, join a tab, or touch the chain
Tab member	View the tab. Log, edit and delete expenses. Run netting. Settle their own payments. Link their own wallet	View other tabs. Edit settlement records. Move another member's funds
Tab creator	Everything a member can do, plus invite and remove members with a zero balance	Remove a member with a non-zero balance. Move another member's funds
System (server and paymaster)	Validate tokens, store tab state, sponsor gas within limits	Custody user funds or sign on a user's behalf

16. Glossary

Term	Meaning
Tab	A shared ledger for one group of members and their expenses
Net position	What a member is owed (positive) or owes (negative) after all expenses
Netting	Combining all net positions into a short list of payments that settles everyone
Per-currency baseline	The payments needed if each currency is netted separately, used only as a comparison

Term	Meaning
Seed rate / indicative rate	A fixed or hardcoded exchange rate, labeled as an estimate and not live
Live rate	A rate read from the Pyth Hermes price feed at the time of entry
Sample mode	A local simulation that never writes on-chain
Testnet mode	Real transactions on Monad Testnet using test tokens
Passkey	Sign-in with face or fingerprint instead of a password
Authenticated session	The period after sign-in. On devices without an enrolled passkey, payments inside it are signed after a slide or typed-amount confirmation, with no biometric prompt
Smart account	A blockchain account created automatically for each user, controlled by their passkey
User operation	A bundled instruction that a smart account sends. It can include several steps
Paymaster	A service that pays network fees on behalf of the user
Finality	The point at which a transaction can no longer be reversed. Monad's published figure is 800 ms
Testnet AUSD	The issuer's official testnet token (6 decimals) used for settlement. It has no value. The issuer's faucet provides it

17. Change Log

Version	Date	Change
1.0	Oct 2026	Initial PRD
1.1	Oct 2026	Aligned to verified facts: mock token named, speed claims corrected, personas unified, claims reworded
2.4	Oct 2026	Aligned to the developer's final code reading: one operation for all creditors, device confirmation by biometric or slide/typed amount, 8-character join codes, automatic wallet registration, three workspace modes plus Developer toggle, in-app nudge, simulated Cash Out and Card, property-test result, contract facts (plaintext tab identifier, HINCRBYFLOAT), live-app transaction hashes
2.3	Oct 2026	Settlement batched into one user operation and measured: median 1.28 s over 10 runs. In-session signing without a further biometric prompt disclosed. Sponsorship guardrails documented (whitelist, monthly ceiling, no per-user cap). Verification items updated. Screenshot removed
2.2	Oct 2026	Token decision changed: official testnet AUSD only, mock token retired. Faucet limits corrected after an on-chain re-read (10,000 per claim, 100,000 ceiling, 60 s). Team names added. Sign-off completed
2.1	Oct 2026	Capacity set (15 supported, 20 ceiling). Roadmap wording for security review, fiat funding and card. Sign-off reduced to two names. Appendix B extended (S9 to S11)
2.0	Oct 2026	Rebuilt at benchmark depth: 10 personas in 3 tiers, 18 stories with measurable criteria, traceability matrix, NFR and edge-case expansion, verification register, risk register, judge self-test path. Figures made illustrative. Other products referenced by category in the body, with named evidence in Appendix B

18. Sign-off

Role / Responsibility	Designated Signatory	Date	Status
Co-founder and Product Lead	Osasumwen Ngozi Scott (ONSInsights)	Oct 2026	Pending
Co-founder and Developer (QA and verification shared)	Miracle Aigbomian (miracles2motion)	Oct 2026	Pending

Appendix A: Worked Example (Illustrative)

This is one example, not a promise. Counts and amounts change with the number of members, the number of currencies, who paid and how costs were split. The final demo seed may differ (V-01). The properties in Section 1 hold for any tab.

Setup. Four members (Aisha, Kemi, Ben, You), six expenses, equal four-way splits, seed rates of 1.09 for EUR and 1.28 for GBP.

Expense	Paid by	Original	Seed rate	USD (cents)
Villa	Aisha	EUR 550	1.09	$599.50 (59,950)
Dinners and wine	Kemi	EUR 240	1.09	$261.60 (26,160)
Excursion	Ben	EUR 200	1.09	$218.00 (21,800)
Flight upgrade	Kemi	GBP 125	1.28	$160.00 (16,000)
Airport transfers	Aisha	USD 120	1.00	$120.00 (12,000)
Sunset cruise	Ben	USD 140	1.00	$140.00 (14,000)

Total spend: **$1,499.10** (149,910 cents). Remainder cents on the villa (59,950 does not divide by 4) are assigned deterministically to the first two members in order.

Member	Net position	Netted payment
Aisha	+$344.72	Receives $16.77 from Ben and $327.95 from You
Kemi	+$46.82	Receives $46.82 from You
Ben	-$16.77	Pays Aisha $16.77
You	-$374.77	Pays Aisha $327.95 and Kemi $46.82

Result. 3 payments in one currency. Check: positive balances total $391.54, negative balances total $391.54, and the three payments total $391.54. Under per-currency netting this tab would need 9 payments across the three currencies. The app computes this baseline itself with its per-currency netting function.

What changes with other inputs. With one currency, the payment count equals what any optimal netting gives and MeshTab's advantage is one-tap settlement. With more currencies or members, the per-currency baseline can grow while MeshTab's list stays within (non-zero members minus 1).

Appendix B: Competitive Evidence (Team Reference)

For internal verification. The body of this PRD refers to other products by category (Web2 expense trackers, Web2 payment apps, Web3 payment apps). This appendix names the sources so each category-level statement can be checked. Remove or keep per the submission format. Re-verify before publishing.

Ref	Statement used in this PRD	Source (accessed Oct 2026)
S1	76% of Gen Z who fronted group costs were not fully repaid. 14% of Gen Z and 19% of Millennials ended a relationship over repayment (US sample)	Zelle Avoidance Economy report
S2	Debt simplification works within each currency separately. It is a free feature that any member can toggle. Pro adds receipt scanning, currency conversion, unlimited expenses, charts	Splitwise help center and Pro page
S3	In-app payments exist only in the US (bank account, SSN required) and India (Paytm integration). Offline entry is supported	Splitwise Pay page and App Store listing

Ref	Statement used in this PRD	Source (accessed Oct 2026)
S4	Another free tracker supports multiple currencies, offline use, receipt photo attachment, payment requests and debt simplification	Tricount product pages
S5	A neobank's group bills feature is free. Non-customers join through an email-verified web profile and get no payment services	Revolut Group Bills help pages
S6	A multi-currency account provider offers 40+ currencies at the mid-market rate, with fees from about 0.23%. This limits any claim about FX loss to banks and cards	Wise product pages
S7	Monad has 400 ms blocks and 800 ms finality. Testnet chain ID 10143. Mainnet chain ID 143, launched November 24, 2025	Monad developer documentation and network pages
S8	Metropolis runs Sept 1 to Oct 13, 2026, with a payments track describing an app that never mentions a blockchain	Monad Metropolis event pages
S9	The issuer publishes a Monad testnet AUSD (0xa9012a055bd4e0eDfF8Ce09f960291C09D5322dC) and a faucet contract (0xd236c18D274E54FAccC3dd9DDA4b27965a73ee6C). On-chain state read by the team: drip 10,000 AUSD, wallet ceiling 100,000 AUSD, one claim per 60 s. Evidence: claim transaction 0x22dc9d3e…2e04, script-executed proof settlement 0xafd413d0…29f0 and live-app batched settlement 0xb3e1987d…58c1 (V-16)	Agora contract-deployments page, Monadscan
S10	Privy funding: card on-ramp through Stripe Crypto Onramp and other providers, bank-transfer on- and off-ramps through Bridge with KYC. Card program built on Bridge and Stripe Issuing, with guided onboarding. Stripe lists stablecoin card issuing as private preview. Network and asset coverage is not confirmed	Privy documentation, Stripe documentation
S11	ack3 sponsors a security scan for every winning team (a scan, not an audit, and only for winners)	Monad Metropolis official page